Department of Information and Communication Technology
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Software Requirement Specification (SRS)

FacultyHub

Object Oriented Programming Practicum
ICT2132

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1. Introduction

1.1 Purpose

The purpose of the **FacultyHub Management Information System (MIS)** is to provide a centralized software solution to manage academic and administrative activities within the Faculty of Technology.

The system enables secure and efficient management of:

- User profiles and role management
- Course information
- Undergraduate attendance records
- Examination marks and grading
- Medical submissions
- Academic notices
- Timetables

FacultyHub improves operational efficiency, minimizes manual paperwork, and ensures accurate academic record maintenance.

1.2 Scope

FacultyHub is a **role-based desktop application** developed using a client–server architecture consisting of:

- **JavaFX Desktop Application (Client / Frontend)**
- **Java Socket Server (Backend)**
- **MySQL Database (Data Storage)**

The system supports the following user roles:

- Administrator
- Lecturer
- Technical Officer
- Undergraduate

All user operations are processed through a centralized server to ensure secure and controlled data access.

1.3 System Overview

FacultyHub provides:

- Secure authentication using JWT tokens

- Role-based authorization
- Academic record management
- Attendance monitoring
- Marks and GPA calculation
- Notice and timetable management

The system follows **Object-Oriented Programming principles** and a **Client–Server Architecture** to ensure modularity, scalability, and maintainability.

2. Overall Description

2.1 System Objectives

Efficient User Management

- Role-based login system
- Secure authentication and authorization

Academic Automation

- Automatic grade calculation
- Attendance eligibility checking
- GPA and CGPA computation

Data Integrity and Security

- Encrypted authentication mechanisms
- Controlled access to academic data

Operational Efficiency

- Reduction of manual processes
- Faster data retrieval and reporting

2.2 System Functions

User Profile Management

Users can:

- View profile information
- Update permitted personal details

Course Management

- Admin creates and manages courses
- Lecturers are assigned to courses
- Students can view enrolled courses

Marks Management

- Lecturers upload and update marks
- Continuous Assessment (CA) validation
- Automatic grade calculation
- Eligibility checking for final exams

Attendance Management

- Technical Officers record attendance
- Theory and practical attendance tracked separately
- Attendance summaries available per student and batch

Medical Management

- Medical records entered and maintained
- Medical approval or rejection tracking
- Attendance adjustments based on approved medicals

Notice Management

- Admin and Lecturers publish notices
- All users can view notices

Timetable Management

- Admin manages timetables
- Users view role-specific timetables

2.3 User Classes and Characteristics

Feature	Admin	Lecturer	Student	Technical Officer
Manage Users	✅	❌	❌	❌
Manage Courses	✅	❌	❌	❌
Upload Materials	❌	✅	❌	❌
Enter Marks	❌	✅	❌	❌
View Grades	✅	✅	✅	❌
Manage Attendance	❌	❌	❌	✅
Manage Medicals	✅	❌	❌	✅
View Notices	✅	✅	✅	✅
View Timetables	✅	✅	✅	✅

2.4 Operating Environment

Hardware Requirements

- CPU: Dual-Core or higher
- RAM: 8GB recommended
- Storage: Minimum 50GB

Software Requirements

- Operating System: Windows or Linux
- Programming Language: Java
- GUI Framework: JavaFX
- Database: MySQL

2.5 Design Constraints

Technology Constraints

- JavaFX must be used for frontend development
- Java Socket programming for backend communication
- MySQL as the database system

Security Constraints

- JWT-based authentication
- Role-based authorization
- Secure password hashing

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Administrator

- Create and manage user accounts
- Assign user roles
- Manage courses and timetables
- Publish notices
- Approve or reject medical submissions
- Generate academic reports

3.1.2 Lecturer

- Update profile information (except credentials)
- Upload course materials
- Enter and update examination marks
- View student academic details
- Check eligibility status
- Publish notices

3.1.3 Technical Officer

- Maintain undergraduate attendance
- Record medical details
- View department timetables
- Assist academic operations

3.1.4 Undergraduate

- View enrolled courses
- View attendance records
- View marks, grades, SGPA, and CGPA
- Submit medical records
- View notices and timetables

- Update contact details and profile picture

3.1.5 Authentication System

- Users log in using username and password
- Server generates JWT tokens
- Token required for every request
- Token validation performed at server

3.2 Non-Functional Requirements

Performance

- Authentication response time < 5 seconds
- Database query response < 2 seconds

Security

- JWT token validation
- Password hashing
- Secure socket communication

Usability

- User-friendly navigation
- Clear validation and error messages
- Learnable within 5 minutes

Scalability

- Future support for mobile and web clients

4. Performance Requirements

- Support concurrent multi-user access
- Handle multiple socket connections simultaneously
- Maintain fast server response times

5. Security Requirements

- JWT-based authentication
- Role-based access control
- Token verification for each request
- Secure database connectivity

6. External Interface Requirements

6.1 User Interface

- JavaFX graphical interface
- Dashboard-based navigation
- Role-specific menus

6.2 Software Interfaces

- MySQL database integration
- JSON-based communication

Example Request

```
{
  "command": "GET_ALL_USERS",
  "token": "JWT_TOKEN",
  "data": {}
}
```

6.3 Communication Interface

- TCP Socket communication
- JSON Request–Response protocol

7. Additional Requirements

7.1 Accessibility

- Keyboard navigation support
- Tooltips and validation messages
- Clear UI feedback

8. System Architecture

Client (JavaFX Desktop Application)
↓ JSON Request via Socket
Server (Command → Service → DAO Layer)
↓ SQL Queries
Database (MySQL)

9. References

- MySQL Official Documentation
- Java SE Documentation
- JavaFX Documentation
- JWT (JSON Web Token) Documentation